A COMPLEX IS PSEUDO-COHERENT EXACTLY WHEN IT IS M-PSEUDO-COHERENT FOR ALL M, AND THEN ITS FINITE FREE RESOLUTION CAN BE CHOSEN TO VANISH ABOVE THE TOP COHOMOLOGY

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Editorial note. The proof uses Xy-pic only through the project's own xymatrix macro, supplied by the retained preamble. The same-chapter text immediately preceding the counted span contains that bibliography citation and is deliberately kept outside both counted spans.

Allowed standard prerequisites (editorial).

- The chapter's own definition of an m-pseudo-coherent and of a pseudo-coherent object of D(R), and of the same terms for R-modules. Definition definition-pseudo-coherent at source lines 15873-15893 of the same chapter, referred to by the text at source line 15903. Fixes what the three listed conditions mean: condition (2) asks for m-pseudo-coherence for every m in Z, condition (3) is the reformulation in terms of a bounded above complex of finite projective modules, and the last sentence of the statement refines the resolution that (3) produces.
- Bounded complexes of finite free or finite projective R-modules and the stupid truncation sigma_{>= m} of a complex. used at proof lines 16064-16076 and 16079-16084. Supplies the objects the three conditions speak about and the truncation that turns a bounded above finite free resolution into a witness of m-pseudo-coherence for a single m.

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- The cone of a morphism of complexes of R-modules together with the long exact cohomology sequence of the resulting distinguished triangle. used at proof lines 16102-16112, where the cone is the one of Derived Categories, Definition derived-definition-cone. Converts the two prescribed cohomological properties of the partially built map alpha into the vanishing of H^i(C^bullet) for i >= n, which is what lets the induction step be restarted one degree lower.

Source reference. Original definition 014E: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1469–1469. Cross-chapter pointer into the Derived Categories chapter fixing the cone of a morphism of complexes; the induction step of the counted proof builds that cone itself and every cohomological conclusion drawn from it stays inside the counted proof span.

Source reference. Original lemma 064R: sources/stacks-more-algebra/pinned-original-source.tex, source lines 15910–15910. Same-chapter pointer used once at proof line 16110 to transfer pseudo-coherence from the two outer terms of a distinguished triangle to the cone; it is one application of a standard two-out-of-three statement and carries no part of the induction that is not written out in the counted span.

Source reference. Original lemma 064S: sources/stacks-more-algebra/pinned-original-source.tex, source lines 15965–15965. Same-chapter pointer used once at proof line 16111 to conclude that a single cohomology module of the cone is a finite R-module; the counted proof then exhibits the surjection onto that module explicitly.

Lemma 0.2. *Let R be a ring. Let $K^\bullet$ be a complex of R-modules. The following are equivalent*

(1) *$K^\bullet$ is pseudo-coherent,*
(2) *$K^\bullet$ is m-pseudo-coherent for every $m \in \mathbf{Z}$, and*
(3) *$K^\bullet$ is quasi-isomorphic to a bounded above complex of finite projective R-modules.*

If (1), (2), and (3) hold and $H^i(K^\bullet) = 0$ for $i > b$, then we can find a quasi-isomorphism $F^\bullet \to K^\bullet$ with F^i finite free R-modules and $F^i = 0$ for $i > b$.

Proof. We see that (1) $\Rightarrow$ (3) as a finite free module is a finite projective R-module. Conversely, suppose $P^\bullet$ is a bounded above complex of finite projective R-modules. Say $P^i = 0$ for $i > n_0$. We choose a direct sum decompositions $F^{n_0} = P^{n_0} \oplus C^{n_0}$ with F^{n_0} a finite free R-module, and inductively

$$F^{n-1} = P^{n-1} \oplus C^n \oplus C^{n-1}$$

for $n \leq n_0$ with F^{n_0} a finite free R-module. As a complex $F^\bullet$ has maps $F^{n-1} \to F^n$ which agree with $P^{n-1} \to P^n$, induce the identity $C^n \to C^n$, and are zero on C^{n-1}. The map $F^\bullet \to P^\bullet$ is a quasi-isomorphism (even a homotopy equivalence) and hence (3) implies (1).

Assume (1). Let $E^\bullet$ be a bounded above complex of finite free R-modules and let $E^\bullet \to K^\bullet$ be a quasi-isomorphism. Then the induced maps $\sigma_{\geq m} E^\bullet \to K^\bullet$ from the stupid truncation of $E^\bullet$ to $K^\bullet$ show that $K^\bullet$ is m-pseudo-coherent. Hence (1) implies (2).

Assume (2). Since $K^\bullet$ is 0-pseudo-coherent we see in particular that $K^\bullet$ is bounded above. Let b be an integer such that $H^i(K^\bullet) = 0$ for $i > b$. By descending induction on $n \in \mathbf{Z}$ we are going to construct finite free R-modules F^i for $i \geq n$, differentials $d^i : F^i \to F^{i+1}$ for $i \geq n$, maps $\alpha : F^i \to K^i$ compatible with differentials, such

that (1) $H^i(\alpha)$ is an isomorphism for $i > n$ and surjective for $i = n$, and (2) $F^i = 0$ for $i > b$. Picture

$$\begin{array}{ccccccc} & & F^n & \longrightarrow & F^{n+1} & \longrightarrow & \ldots \\ & & \downarrow{\scriptstyle \alpha} & & \downarrow{\scriptstyle \alpha} & & \\ K^{n-1} & \longrightarrow & K^n & \longrightarrow & K^{n+1} & \longrightarrow & \ldots \end{array}$$

The base case is $n = b + 1$ where we can take $F^i = 0$ for all i. Induction step. Let $C^\bullet$ be the cone on α (Derived Categories, Definition 014E). The long exact sequence of cohomology

$$0 \to H^{n-1}(K^\bullet) \to H^{n-1}(C^\bullet) \to H^n(F^\bullet) \to H^n(K^\bullet) \to H^n(C^\bullet) \to \ldots$$

shows that $H^i(C^\bullet) = 0$ for $i \geq n$. By Lemma 064R we see that $C^\bullet$ is $(n-1)$-pseudo-coherent. By Lemma 064S we see that $H^{n-1}(C^\bullet)$ is a finite R-module. In particular, we see that the kernel of $H^n(F^\bullet) \to H^n(K^\bullet)$ is a finite R-module. Choose a finite free R-module F^{n-1} and a map $F^{n-1} \to \mathrm{Ker}(F^n \to F^{n-1})$ such that F^{n-1} surjects onto the kernel of $H^n(F^\bullet) \to H^n(K^\bullet)$. We extend our of complexes to

$$\begin{array}{ccccccccc} & & F^{n-1} & \longrightarrow & F^n & \longrightarrow & F^{n+1} & \longrightarrow & \ldots \\ & & \downarrow{\scriptstyle \alpha^{n-1}} & & \downarrow{\scriptstyle \alpha} & & \downarrow{\scriptstyle \alpha} & & \\ \ldots & \longrightarrow & K^{n-1} & \longrightarrow & K^n & \longrightarrow & K^{n+1} & \longrightarrow & \ldots \end{array}$$

Note that α^{n-1} exists because the image of $F^{n-1} \to F^n \to K^n$ is in the image of $K^{n-1} \to K^n$ by construction. Dnote again $C^\bullet$ the cone of this extended map of complexes. At this point we see that we get an exact sequence

$$H^{n-1}(F^\bullet) \to H^{n-1}(K^\bullet) \to H^{n-1}(C^\bullet) \to H^n(F^\bullet) \cong H^n(K^\bullet) \to H^n(C^\bullet) \to \ldots$$

In other words, we see that the cokernel of $H^{n-1}(F^\bullet) \to H^{n-1}(K^\bullet)$ is a finite R-module, say generated by the classes of $\xi_1, \ldots, \xi_r \in \mathrm{Ker}(K^{n-1} \to K^n)$. Then we replace F^{n-1} by $F^{n-1} \oplus R^{\oplus r}$ where the basis elements in the free summand map to zero in F^n and the to ξ_i in K^{n-1}. This finishes the proof of the induction step. $\square$